

Active Layer Depth and Permafrost Temperatures at the Teller 47 Field Site, Seward Peninsula, Alaska, 2022

NGEE Arctic Record_id: NGA544



Summary:

With extreme climate warming in the Arctic, there is increasing focus on mapping permafrost stability and improving understanding of the impacts of thawing permafrost. Summer active layer depths and temperatures at the top of the permafrost were collected at the Teller 47 field site on the Seward Peninsula, Alaska to better understand permafrost thermal state in areas of discontinuous permafrost. Measurements were taken between August 14th and August 20th, 2022. Active layer depths were measured with a 120 cm thaw probe, and temperature measurements were taken with a SpotOn Temperature probe at the top of permafrost and at a maximum depth of XYZ, when no near-surface permafrost (within 120 cm of the surface) was present. Measurements were co-located with ground surface displacement measurements, which can be found in the associated dataset, NGA254. Precise location data were collected at each observation point using a GNSS RTK GPS unit. This dataset contains a *.csv file of permafrost temperature and active layer depths and a *.kml file of measurement locations.

Please use this citation to reference the data.

Evan Thaler, Joanmarie Del Vecchio, Margaret Farley, Lauren Thomas, Joel Rowland. 2024 Active Layer Depth and Permafrost Temperatures at the Teller 47 Field Site, Seward Peninsula, Alaska, 2022. Next Generation Ecosystem Experiments Arctic Data Collection, Oak Ridge National Laboratory, U.S. Department of Energy, Oak Ridge, Tennessee, USA. Dataset accessed on [INSERT_DATE] at [DOI].

NGEE Arctic Project Summary

The Next-Generation Ecosystem Experiments: Arctic (NGEE Arctic), was a research effort to reduce uncertainty in Earth System Models by developing a predictive understanding of carbon-rich Arctic ecosystems and feedbacks to climate. NGEE Arctic was supported by the Department of Energy's Office of Biological and Environmental Research.

The NGEE Arctic project had two field research sites: 1) located within the Arctic polygonal tundra coastal region on the Barrow Environmental Observatory (BEO) and the North Slope near Utqiagvik

(Barrow), Alaska and 2) multiple areas on the discontinuous permafrost region of the Seward Peninsula north of Nome, Alaska.

Through observations, experiments, and synthesis with existing datasets, NGEE Arctic provided an enhanced knowledge base for multi-scale modeling and contributed to improved process representation at global pan-Arctic scales within the Department of Energy's Earth system Model (the Energy Exascale Earth System Model, or E3SM), and specifically within the E3SM Land Model component (ELM).

Data Acquisition Materials and Methods

Between August 14th – 20th in 2022, temperature at top of permafrost was measured at previously established survey locations, which can be found in NGA254 (Lathrop et al. 2022). Temperatures were measured with a SpotOn temperature probe, which has an accuracy of 0.8°C, and was calibrated to 0°C in ice water. Temperature measurements were collected to a depth of 0.75 – 1.20 m depth. At each temperature measurement location, active layer depth was also measured using a soil/frost probe. A measurement of 120 cm indicates that no near-surface permafrost was detected. Precise location data were collected at each observation point using a GNSS RTK GPS unit.



Figure 1. Measurement probes at the Teller 47 field site. Left, a SpotOn temperature probe. Right, a frost probe used to measure active layer depth.

References

SpotOn Temperature Probe: <https://innoquestinc.com/product/spoton-temp-probe/>

GNSS RTK GPS unit: Trimble dGPS <https://geospatial.trimble.com/en/products/hardware/gnss-systems>

Lathrop, E., Rowland, J., DelVecchio, J., Charsley-Groffman, L., Thaler, E., Swanson, E., Dann, J., Sutfin, N. 2022. DGPS measurements of solifluction and hillslopes creep at the Teller 47 field site from 2017-2019, Seward Peninsula, AK. Next Generation Ecosystem Experiments Arctic Data Collection, Oak Ridge National Laboratory, U.S. Department of Energy, Oak Ridge, Tennessee, USA.

<https://doi.org/10.5440/1881334>. <https://data.ngee-arctic.ornl.gov/data/pages/NGA254.html>.

Thaler, E. A., Uhleman, S., Rowland, J. C., Schwenk, J., Wang, C., Dafflon, B., & Bennett, K. E. 2023. High-Resolution Maps of Near-Surface Permafrost for Three Watersheds on the Seward Peninsula, Alaska Derived From Machine Learning. *Earth and Space Science*, 10(12), e2023EA003015.

<https://doi.org/10.1029/2023EA003015>

Supplemental Files

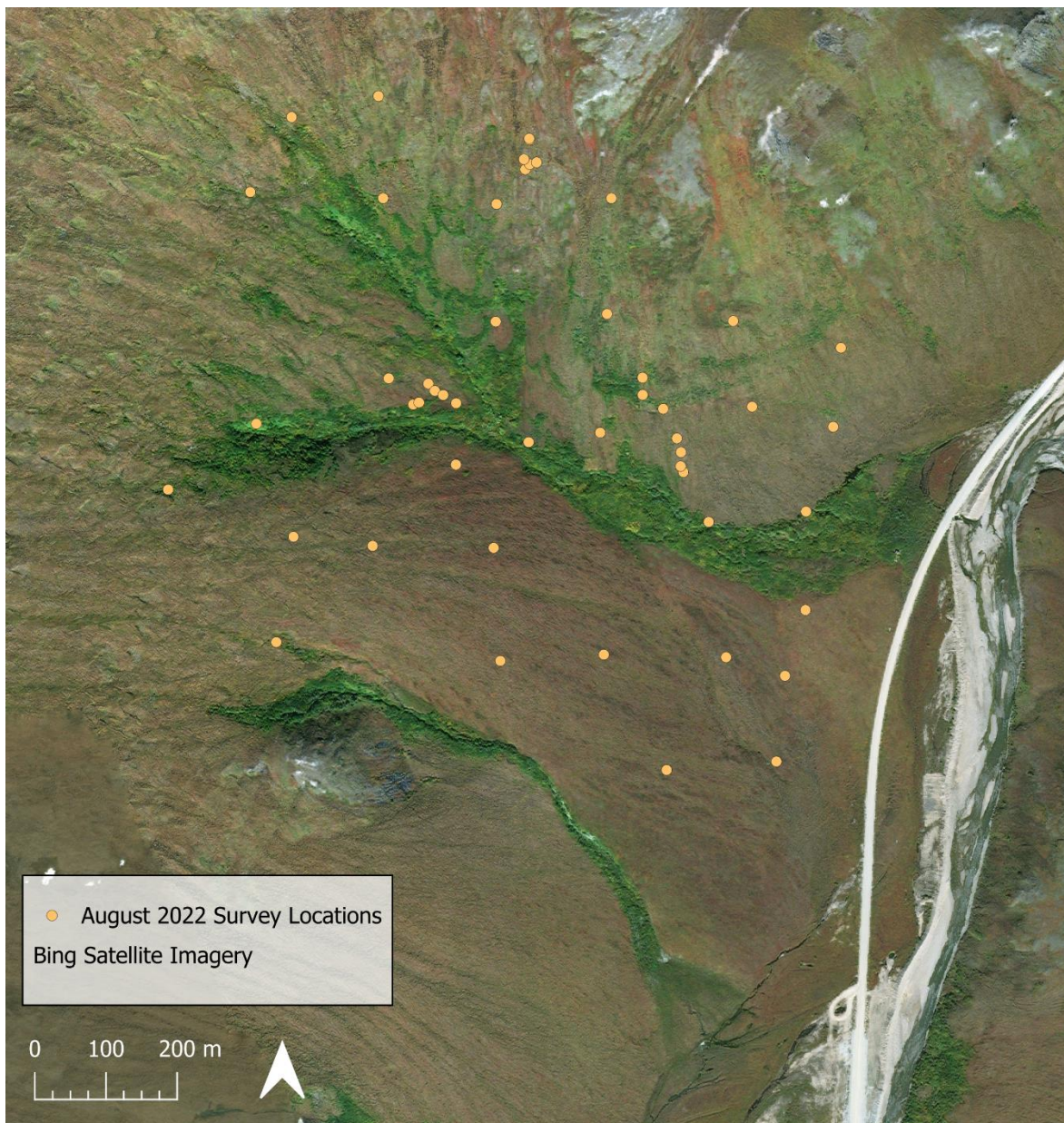


Figure 2: August 2022 survey locations at Teller 47 site.